

Prospective studies of dietary alpha-linolenic acid intake and prostate cancer risk: a meta-analysis.



Individual-based studies that investigated the relation between dietary alpha-linolenic acid (ALA) intake and prostate cancer risk have shown inconsistent results. We carried out a meta-analysis of prospective studies to examine this association. We systematically searched studies published up to December 2008. Log relative risks (RRs) were weighted by the inverse of their variances to obtain a pooled estimate with its 95% confidence interval (CI). We identified five prospective studies that met our inclusion criteria and reported risk estimates by categories of ALA intake. Comparing the highest to the lowest ALA intake category, the pooled RR was 0.97 (95% CI:0.86-1.10) but the association was heterogeneous. Using the reported numbers of cases and non-cases in each category of ALA intake, we found that subjects who consumed more than 1.5 g/day of ALA compared with subjects who consumed less than 1.5 g/day had a significant decreased risk of prostate cancer: RR = 0.95 (95% CI:0.91-0.99). Divergences in results could partly be explained by differences in sample sizes and adjustment but they also highlight limits in dietary ALA assessment in such prospective studies. Our findings support a weak protective association between dietary ALA intake and prostate cancer risk but further research is needed to conclude on this question.